🤌 XAI Lecture 14

Network Dissection & TCAV

FAMAF - UNC

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eXplainable
Artificial Intelligence

🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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https://interpretable-ml-class.github.io/

🧉 Paper 1 ---

Network Dissection: Quantifying Interpretability of Deep Visual Representations

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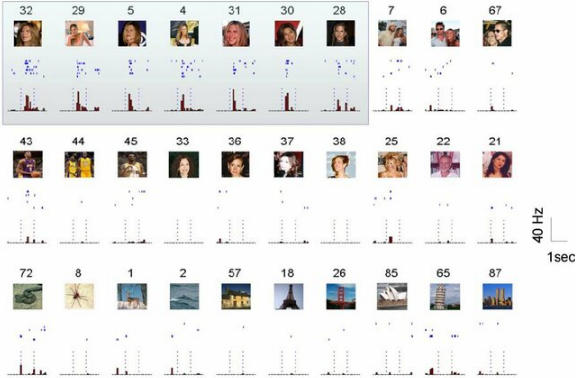
Presentation Roadmap ---

- **Introduction**
  - Motivation
  - Questions the paper aims to answer
  - Related works
- **Method**
- **Experiments**
  - Training Conditions
  - Discrimination
  - Layer Width

🧉 How Is Semantic Visual Concept Represented in the Brain? ---



Neuroscientists have found that the brain uses sparse, localized representations: individual neurons (or small groups) respond selectively to specific concepts.

Key Observation

Some neurons fire strongly for one concept and weakly for almost everything else — a property called **selectivity**.

Question: Do deep neural networks learn similar structure?

🧉 A Neuron That Only Fires for Jennifer Aniston ---

Quiroga et al. (2005) recorded neurons in the human medial temporal lobe and found neurons that fire exclusively for specific people or landmarks.

- One neuron fired for Jennifer Aniston — photos, drawings, even her name written in text
- Another neuron fired only for the Eiffel Tower

Why this matters

This is a **disentangled** representation: one neuron = one concept. The paper asks whether CNNs learn something analogous.

🧉 Disentangled Representation in Visual Cortex ---

The visual cortex processes information in a hierarchy of layers:

- V1: edges and orientations
- V2/V4: textures, colors, shapes
- IT (inferotemporal cortex): objects and faces

Each area builds on the previous one — low-level features combine into high-level concepts.

Analogy

This layered specialization is the biological inspiration for deep CNNs. Do CNNs also develop specialized detectors at each layer?

Deep CNN for Computer Vision ---

CNNs revolutionized computer vision, reaching superhuman performance on ImageNet — but they remain largely opaque.

- Units in early layers respond to edges and colors
- Units in later layers seem to respond to objects and scenes (Zhou et al., 2015)
- But this was observed qualitatively — no rigorous measurement existed

The gap

We know CNNs *work* extremely well. We don't know *why*, or what exactly each unit has learned.

🧉 Proposed Questions ---

The paper frames the problem around three concrete research questions. Each has a direct answer by the end of the paper.

1. **What is a disentangled representation, and how can its factors be quantified and detected (in deep CNN)?**
2. **Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?**
3. **What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?**

🧉 Proposed Questions and Contributions ---

What is a disentangled representation, and how can its factors be quantified and detected?

- Proposed a metric, intersection over union score (IoU), to quantify the interpretability of each unit
- The alignment level between unit activated area and human-interpretable concepts

Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?

- A semantic concept can be detected by many units
- A unit can detect many semantic concepts

What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?

- Number of unique detectors, layer depth, training iterations
- The angle of the images, input datasets
- Fine-tuning, supervised vs. unsupervised

🧉 Related Works ---

Prior work tried to understand CNN internals through **visualization**, but all approaches share a fundamental limitation: they are qualitative and cannot be used to rigorously compare models.

1. **Generative Visualizations of Individual Units**
   - Synthesize images that maximally activate a unit
   - Mahendran et al., CVPR 2015; Nguyen et al., NIPS 2016; Simonyan et al., ICML 2014
2. **Salience-based Visualizations of Individual Units**
   - Highlight which pixels most contribute to a unit's activation
   - Deconvolution: Zeiler et al., ECCV 2014
3. **Visualizing Representations as a Whole**
   - Project the full representation space into 2D for inspection
   - t-SNE: Maaten et al., JMLR, 2008; Yosinski et al., ICML, 2015

Common limitation

All these methods produce images that **humans** must then interpret — subjective and impossible to compare across networks.

🧉 Related Works: The Need for Quantification ---

Prior work focused on **visualizing** internals, but these methods remain qualitative and subjective.

1. **Generative Visualizations**: Synthesizing images that maximize activation (e.g., Activation Maximization).
2. **Salience-based Methods**: Highlighting pixels that contribute most to a prediction (e.g., Deconvolution).
3. **Global Analysis**: Visualizing the representation space as a whole (e.g., t-SNE).

The Semantic Gap

Visualizations still require a **human** to interpret the resulting images. Network Dissection aims to automate this by matching units with labeled concepts directly.

🧉 Method ---

Broden: Broadly and Densely Labeled Dataset ---

To measure interpretability we need a **ground truth** of human concepts. Broden unifies five existing segmentation datasets into a single resource.

broden dataset

Category	# Classes
Scene	468
Object	584
Part	234
Material	32
Texture	47
Color	11

Multiple labels can apply to the **same pixel** (e.g., a black cat leg → “cat”, “leg”, “black”)

🧉 Scoring Unit Interpretability: The Process ---

The method evaluates every convolutional unit k as a solution to a **binary segmentation task** for every concept c.

1. **Activation Collection**: Extract the activation map $A_k(x)$ from the frozen network.
2. **Quantile Thresholding**: Define a threshold T_k such that $P(a_k > T_k) = 0.005$ over the dataset. This identifies the "top" activations.
3. **Resolution Alignment**: Since $A_k(x)$ is lower resolution, use **bilinear interpolation** to upsample it to $S_k(x)$.
4. **Binary Mask**: Generate $M_k(x) \equiv S_k(x) \geq T_k$, selecting regions where the unit is "active".

This is a direct probe; no training or backpropagation is required.

🧉 Scoring Unit Interpretability ---

Each convolutional unit k acts like a **filter** that fires on certain spatial regions of an image. The core idea: treat each unit as a candidate **segmentation model** for some concept.

activation map

Pipeline (no retraining needed):

1. Feed images from Broden through the frozen CNN
2. Collect activation map $A_k(\mathbf{x})$ for every unit k
3. Upsample $A_k(\mathbf{x})$ to input resolution
4. Threshold to obtain a binary mask $M_k(\mathbf{x})$
5. Compare $M_k(\mathbf{x})$ against concept labels $L_c(\mathbf{x})$

🧉 Scoring Unit Interpretability: Full Pipeline ---

scoring pipeline

For each unit, the method evaluates its activation mask against all **1,197 segmentation tasks** in Broden (one per concept). The unit is assigned the concept with the highest score.

🧉 Scoring Unit Interpretability ---

scoring threshold

Threshold T_k: value m such that $P(a_k^{ij} > m) = 0.005$

🧉 Scoring Unit Interpretability ---

scoring iou

$$IoU_{k,c} = \frac{\sum |M_k(\mathbf{x}) \cap L_c(\mathbf{x})|}{\sum |M_k(\mathbf{x}) \cup L_c(\mathbf{x})|}$$

🧉 Quantifying Alignment: The IoU Score ---

The interpretability of unit k for concept c is the **Intersection over Union (IoU)** across the dataset:

$$IoU_{k,c} = \frac{\sum |M_k(x) \cap L_c(x)|}{\sum |M_k(x) \cup L_c(x)|}$$

- **Detector Definition**: A unit k is considered a detector for concept c if $IoU_{k,c} > 0.04$.
- **Top Label**: If a unit matches multiple concepts, the top-ranked label is assigned.
- **Layer Score**: The interpretability of a layer is the number of **unique** semantic concepts aligned with its units.

🧉 Scoring Unit Interpretability ---

$$IoU_{k,c} = \frac{\sum |M_k(\mathbf{x}) \cap L_c(\mathbf{x})|}{\sum |M_k(\mathbf{x}) \cup L_c(\mathbf{x})|}$$

- Changing *IoU* threshold changes number of concept detectors but not orderings between networks
- One unit might be a detector for multiple concepts; they choose the top-ranked concept for an individual unit
- Interpretability of a layer = number of unique concepts aligned with units

🧉 Experiments ---

🧉 Experiments: Tested CNN Models ---

The paper evaluates interpretability across diverse architectures and training regimes to identify which conditions favor disentangled representations.

Training	Network	Data set or task
none	AlexNet	random
Supervised	AlexNet	ImageNet, Places205, Places365, Hybrid
Supervised	GoogLeNet	ImageNet, Places205, Places365
Supervised	VGG-16	ImageNet, Places205, Places365, Hybrid
Supervised	ResNet-152	ImageNet, Places365
Self	AlexNet	context, puzzle, egomotion, tracking, moving, videoorder, audio, crosschannel, colorization, objectcentric

🧉 Experiment 1: Human Evaluation of Interpretations ---

1. **Identify Interpretable Units** — units that raters agreed with ground-truth interpretations
2. Raters shown 15 images with highlighted patches showing the most highly-activating regions for each unit in AlexNet trained on Places205, and asked to decide (yes/no) whether a given phrase describes most of the image patches
3. **Find Network Dissection** — portion of interpretations generated by method that were rated as descriptive
4. **Human Consistency** — portion of ground-truth labels found descriptive by a second group of raters *exxp1_human_eval*

🧉 Experiment 1: Human Evaluation Results ---

	conv1	conv2	conv3	conv4	conv5
Interpretable units	57/96	126/256	247/384	258/384	194/256
Human consistency	82%	76%	83%	82%	91%
Network Dissection	37%	56%	54%	59%	71%

Network Dissection agreement increases in higher layers; Human consistency remains high throughout.

🧉 Experiment 2: Axis-Aligned Interpretability ---

Two hypotheses:

1. **Concepts appear in every direction**
   - **Default hypothesis:** single units not much more interpretable than combinations of units
2. **Concepts are rare + the model converges to a special, semantically rich basis**
   - The model's natural basis is a meaningful decomposition

Is interpretability an "axis-aligned" property of the natural basis, or do concepts appear in every direction?

- **Method**: Apply a random orthogonal rotation Q to the representation space.
- **Result**: Rotating the basis drastically reduces the number of unique detectors (up to 80% decrease).
- **Conclusion**: Interpretability is **not** an inevitable result of discriminative power. It is a special alignment learned by the network.

🧉 Experiment 2: Axis-Aligned Interpretability ---

exp2_rotation

Number of unique detectors decreases as the basis is rotated — confirming hypothesis 2.

🧉 Experiment 3: Concepts by Layer ---

exp_3

🧉 Experiment 4: Network Architectures ---

exp_4

🧉 Experiment 5: Training Conditions ---

Varied training conditions:

1. **Weight Initializations**
   - **Minimal Effect:** Models converge to similar levels of interpretability
2. **Dropout**
   - **Some Effect:** Lack of dropout leads to more "texture" and less "object" detectors
3. **Batch Normalization**
   - **Significant Effect:** Interpretability decreased significantly

🧉 Experiment 5: Training Conditions ---

exp_5

🧉 Experiment 6: Discrimination vs. Interpretability ---

Does being more "interpretable" make a model better at other tasks?

- **Task**: Use high-level activations to train a linear SVM for *action recognition* (Action40 dataset).
- **Finding**: There is a **positive correlation** between the number of unique object detectors and classification accuracy.
- **Implication**: Encouraging disentangled concept detection can improve the feature's ability to generalize to new tasks.

🧉 Experiment 6: Discrimination ---

Benchmark high-level activations on a new task:

- Across several Deep NNs, extract activations from high CNN layers
- Train a linear SVM on a new *action recognition* task
- Compute classification accuracy

🧉 Experiment 6: Discrimination Results ---

exp_6

Result: Positive correlation between object detectors and classification accuracy $\rightarrow$ encouraging **concept detection** can improve **discrimination**.

🧉 Experiment 7: Width ---

Effect of layer width (number of units in a layer):

Increased layer width retains similar accuracy, but many more **concept detectors**

- # Detectors increased both at the increased layer and in the network generally
- Increase has a threshold

🧉 Experiment 7: Layer Width ---

What happens if we make a layer “wider” (more units)?

- **Method**: Tripled the number of units in AlexNet’s `conv5` (from 256 to 768).
- **Result**: Accuracy remains similar, but the number of **unique concept detectors** increases significantly.
- **Limit**: Increasing width beyond a certain point (e.g., 1024 units) yields diminishing returns in unique concepts.

Wider layers provide more "slots" for the network to separate explanatory factors.

🧉 Experiment 7: Width Results ---

exp_7

🧉 Discussion Questions (Paper 1) ---

1. **Distribution Understanding:** Concept detectors from a particular dataset betray something about the underlying distribution. How do you think this can be applied in the real world (e.g., bias detection)?
2. **Single unit to circuit:** This method interprets single units; could it be extended to larger circuits and would this be useful?
3. **Beyond vision:** This method is deeply tied to the vision domain. Could it be extended to other domains such as natural language?

Thank You!

🧉 References --- |